

SHAHBANO FAROOQ

Ph.D. Candidate, Interactive Arts and Technology
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EDUCATION

M.Sc Computer Science (Jan 2012 – Apr 2014) University of Calgary – Calgary, AB, Canada.

Thesis -Design and Discussion of Visualizations in Pairs: The research explored collaborative visualization design as a mechanism for collective sensemaking and knowledge exchange. The interactive tabletop interface provided four panels for data, concept mapping, visual design and mapping to code generation, to facilitate knowledge exchange and discussions between domain experts and visualization designers.

B.Sc (Honors) Computer Science (Jan 1999 – Dec 2002) Kinnaird College for Women, Lahore, Pakistan.

Final Project: Developed an Urdu dictation system that employed Markov models for speech recognition and language modeling, enabling the conversion of spoken Urdu into text.

AWARDS

PhD Research Scholarship – Simon Fraser University (Fall 2025, Spring 2026, Summer 2026).

Travel and Research Award – Simon Fraser University (Summer 2026).

Graduate Fellowship – Simon Fraser University (Summer 2026).

Queen Elizabeth 2 Scholarship - University of Calgary, Canada (2013).

K.K Aziz Gold Medal - Kinnaird College for Women, Pakistan. (2002)

Outstanding Support - Student Accessibility Center, Zayed university (2023)

RESEARCH EXPERIENCE

Research Assistant, Transforming Arts Impact Initiative (TAIP) (Jan 2026 - To Date): Exploring alternative approaches to empower the art's community in understanding, articulating, and leveraging the impact of their cultural contributions and getting better support from the government and funding organizations.

Personal Data Physicalizations Workshop (May 2026): Participated in a workshop exploring data physicalizations methods by translating personal data into clay-based artistic representations.

Panelist, AI Discussion Panel, FCAT Graduate Forum (April 2025): Participated in a panel discussion examining the opportunities, challenges, and implications of artificial intelligence in higher education.

SFU-UBC Visualization Meetup: Presented ongoing research project on, Generative Tools for the Co-Design of Re-thinking Impact.

Participated in VIVA (Vancouver Institute for Visual Analytics): Participated in and contributed to discussions on current research goals for supporting community based visual analytics research.

TEACHING EXPERIENCE

Instructor, Zayed University, UAE

ICS 220: Programming Fundamentals (Fall 2023, Spring 2024)

ICS 221: Data Structures and Algorithms (Spring 2025)
ICS 340: Database Systems (Spring 2025)
ICS 351: Machine Learning (Fall 2024)
ICB 103: Applied Algorithmic Thinking (Spring 2023, Fall 2024)
SWE 225: Introduction to Programming and Problem Solving (Fall 2022)
SWE 321: Object Oriented Programming Lab (Spring 2021, Fall 2021, Spring 2022, Fall 2022, Spring 2023)
CIT 210: Essentials of IT and Infrastructure (Fall 2020, Spring 2021, Fall 2021, Spring 2019, Fall 2019, Spring 2020, Fall 2020, Spring 2021, Spring 2022, Fall 2022)
GEN 110: Data Management and Analysis (Fall 2019, Fall 2020)
GEN 175: Introduction to Information and Technology (Spring 2017, Fall 2017, Spring 2018, Fall 2018, Internship Supervision: (Spring 2022, Fall 2022)

Teaching Assistant: University of Calgary, Canada

CPSC 203 - Introduction to Problem Solving using Application Software (Fall 2012)

CPSC449 - Programming languages paradigms (Winter 2012)

PUBLICATIONS

JOURNAL ARTICLES

Kuhail, M. A., Al-Shamaileh, O., **Farooq, S.**, Shahin, H., Abdelzaher, F., Thomas, J., & Kaura, N. (2025). *A systematic review on mental health chatbots: Trends, design principles, evaluation methods, and future research agenda. Human Behavior and Emerging Technologies.*

Co-authored a systematic review of 97 studies on mental health chatbots, examining design principles, evaluation methods, therapeutic approaches, usability considerations, and future directions for AI-supported mental health interventions.

Kuhail, M. A., ElSayary, A., **Farooq, S.**, & Alghamdi, A. (2022, September). Exploring immersive learning experiences: A survey. In *Informatics* (Vol. 9, No. 4, p. 75). MDPI.

As a co-author, I investigated the pedagogical impact and design considerations of immersive technologies and interaction modalities and their role in shaping user experience.

Kuhail, M. A., **Farooq, S.**, Hammad, R., & Bahja, M. (2021). Characterizing visual programming approaches for end-user developers: A systematic review. *IEEE Access*, 9, 14181-14202.

I co-investigated how visual interfaces support application development for non-programmers and provided a taxonomy for classification of visual programming interfaces.

BOOK CHAPTERS

Kuhail, M. A., **Farooq, S.**, & Almutairi, S. (2023). Recent Developments in Chatbot Usability and Design Methodologies. *Trends, Applications, and Challenges of Chatbot Technology*, 1-23.

I coauthored a chapter on chatbot usability, which identifies key usability factors applied in chatbot design, various application domains where chatbots have been utilized, and future directions for chatbot design.

Farooq, S., Carpendale, S., & Maurer, F. (2016). *Pairing for designing visualizations*. In F. Maurer (Ed.), *Designing Digital Surface Applications* (pp. 155-165). SurfNet.

This work illustrates how to support knowledge exchange and discussion between a domain expert and a visualization designer during visualization design studies. My work examined interactive interfaces for structuring these collaborative sessions around the role of participants and how interactions can support sensemaking during the design process.

BOOK

Mathew, S. S., Kuhail, M. A., Hadid, M., & **Farooq, S.** (2025). *The Object-Oriented Approach to Problem Solving and Machine Learning with Python*. CRC Press.

I co-authored a book on object-oriented design and development for modular systems, and I solely authored the chapter that introduces machine learning and demonstrates how to leverage object-oriented structures.

THESIS

Farooq, S. (2014). *Design and Discussion of Visualizations in Pairs* (Doctoral dissertation, University of Calgary).

CERTIFICATIONS

Fellowship of the Higher Education Academy - AdvanceHE, London.

Blackboard Academy Certified Associate - Zayed University, UAE.

PRESENTATIONS

UAE Innovates– The Growth Mindset - Zayed University (2025)

PD Session on Power BI - Zayed University (2024)

PD Session for Course Builder on Forum (Minerva) - Zayed university (2023)

Alumni Training on Power BI - Zayed University (2022)

SOTL Conference - Workshop – Improving Student Engagement and Learning with Nearpod (2018)

ALLT Forum - Conference Presentation - Finding Balance in Teaching and Learning (2019)

PD Session on Visualization Design - Zayed University (2018)

COMMITTEES

Teaching and Learning Committee (2024, 2025)

Gulf Programming Contest Committee (2023)

Student Affairs Committee (2022)

Technology Committee (2019, 2020, 2021)

Event Management Committee (2019)

Course Learning Outcomes Committee (2018)

SERVICES

Mentor for the Partner Challenge Projects (2023, 2024)
Mental Health Gate Keeper (2022, 2023)
Chaperon for Expo (2022)
Robotics Challenge Judge – College of Technological Innovation (2022)
Capstone Project Supervisor for High School Students by Ministry of Education (2021)
IEEE Conference Session Chair (2021)
Site Manager - ALLT Conference (2020)
Special Olympics – Record Keeper (2019)
AI Waha Zone – Workshops for At Risk Students. (2017, 2018, 2020)
Smart Gym Event Organizer – (2019, 2020, 2021)

PROFESSIONAL EXPERIENCE

Technical Content Developer at Etisalat - UAE (Spring 2005- Spring 2007)
Software Development and Support Engineer at Emircom – UAE (Fall 2003 – Fall 2004)
System Analyst \ Programmer at ESP GITS – Pakistan (Spring 2002 – Fall 2003)

PROFESSIONAL DEVELOPMENT CERTIFICATES

CENTER OF EDUCATIONAL INNOVATION – ZAYED UNIVERSITY

Enhancing Teaching and Learning through educational technology (2019)
Advancing Academic Integrity (2019)
Evidence-based Approaches to Developing Students' Critical Thinking Skills (2018)
Instructional Design and Planning (2017)
Scholarly Teaching and Learning (2017)
Developing Meaningful Assessments (2017)